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CODE:

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_squared_error, r2_score

# Sample non-linear dataset
X = np.array([
    [1], [2], [3], [4], [5],
    [6], [7], [8], [9], [10]
])

y = np.array([
    2, 5, 10, 17, 26,
    37, 50, 65, 82, 101
])

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ---------------- LINEAR REGRESSION ----------------

linear_model = LinearRegression()

# Train the model
linear_model.fit(X_train, y_train)

# Predict test data
y_pred_linear = linear_model.predict(X_test)

# Calculate performance
mse_linear = mean_squared_error(y_test, y_pred_linear)
r2_linear = r2_score(y_test, y_pred_linear)


# ---------------- POLYNOMIAL REGRESSION ----------------

# Create polynomial features of degree 2
poly = PolynomialFeatures(degree=2)

X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

# Create and train model
poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)

# Predict test data
y_pred_poly = poly_model.predict(X_test_poly)

# Calculate performance
mse_poly = mean_squared_error(y_test, y_pred_poly)
r2_poly = r2_score(y_test, y_pred_poly)
```

```
# ---------------- DISPLAY RESULTS ----------------

print("LINEAR AND POLYNOMIAL REGRESSION COMPARISON")
print("=" * 50)

print("\nLinear Regression:")
print("Mean Squared Error:", round(mse_linear, 4))
print("R-squared Score:", round(r2_linear, 4))

print("\nPolynomial Regression:")
print("Mean Squared Error:", round(mse_poly, 4))
print("R-squared Score:", round(r2_poly, 4))

# Compare models
print("\nMODEL COMPARISON")

if r2_poly > r2_linear:
    print("Polynomial Regression performs better.")
else:
    print("Linear Regression performs better.")

# Predict a new value
new_input = np.array([[11]])

linear_prediction = linear_model.predict(new_input)

poly_prediction = poly_model.predict(
    poly.transform(new_input)
)

print("\nPrediction for X = 11")
print("Linear Regression:", round(linear_prediction[0], 2))
print("Polynomial Regression:", round(poly_prediction[0], 2))
```

OUTPUT :

```
LINEAR AND POLYNOMIAL REGRESSION COMPARISON
==================================================

Linear Regression:
Mean Squared Error: 25.0
R-squared Score: 0.9831

Polynomial Regression:
Mean Squared Error: 0.0
R-squared Score: 1.0

MODEL COMPARISON
Polynomial Regression performs better.

Prediction for X = 11
Linear Regression: 99.0
Polynomial Regression: 122.0
```